

Chapter 4 Homework

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Exercise Group 4.2

- 1) It means that $R = T$ and $S = U$.
- 2) Yes, congruence is reflexive.
- 3) No, two different segments can be congruent to each other.
- 4) Two segments AB and CD are not congruent to each other. Prove that AB and CD are not equal to each other.

Proof. If $AB \not\cong CD$, then $AB \neq CD$.

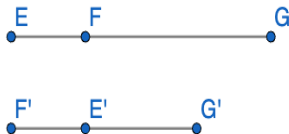
1. We will prove the contrapositive, that is if $AB = CD$ then $AB \cong CD$.
2. Let $AB = CD$, and observe that $AB \cong CD$ since congruence is reflexive. □

- 5) If ABC , then both B and C are on the same side of A . So by III-1, since $AC \cong AC$ then $AC \not\cong AB$.
- 6) It is *false*. There are two points on either side of A' , call them B' and B'' , such that $A'B' \cong AB$ and $A'B'' \cong AB$.
- 7) $DF \cong CB$, $DF \cong AB$, and $ED \cong CD$.
- 10) Observe that since ABC , $A'B'C'$, $AB \cong A'B'$, and $BC \cong B'C'$, then $AC \cong A'C'$ by III-3. Again, since ACD , $A'C'D'$, $AC \cong A'C'$, and $CD \cong C'D'$, then $AD \cong A'D'$ by III-3.
- 11) From the previous exercise, prove that if $AD \cong A'D'$, $CD \cong C'D'$, and $BC \cong B'C'$, then $AB \cong A'B'$.

Proof. If $AD \cong A'D'$, $CD \cong C'D'$, and $BC \cong B'C'$, then $AB \cong A'B'$.

1. Let $AD \cong A'D'$, $CD \cong C'D'$, and $BC \cong B'C'$.
2. Since $AD \cong A'D'$ and $CD \cong C'D'$, then $AC \cong A'C'$ by III-3.
3. Since $AC \cong A'C'$ and $BC \cong B'C'$, then $AB \cong A'B'$ by III-3. □

- 12) $EF \cong E'F'$, $FG \cong F'G'$, but $EG \not\cong E'G'$

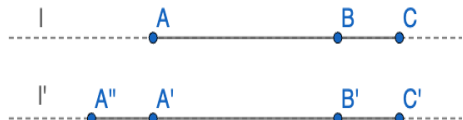


13) Prove part b) of Postulate III-3 using part a).

Proof. If ABC on line l and $A'B'C'$ on l' such that $AC \cong A'C'$ and $BC \cong B'C'$, then $AB \cong A'B'$.

1. Let ABC on line l and $A'B'C'$ on l' such that $AC \cong A'C'$ and $BC \cong B'C'$.
2. Assume BWOC that $AB \not\cong A'B'$.
3. Then there exists $A'' \in l'$ on the same side of B' as A' such that $A''B' \cong AB$.
4. Observe that since we already have $A'C' \cong AC$, then $A''C' \not\cong AC$.
5. Since $A''B' \cong AB$ and $B'C' \cong BC$, then by III-3 we have $A''C' \cong AC \Rightarrow \Leftarrow$.
6. Therefore $AB \cong A'B'$.

□

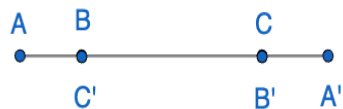


15) a) $BC \cong B'C'$ from III-3 part b)

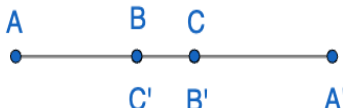
b) $CB \cong AD$ from III-3 part a)

c) $AB \cong CD$ from III-3 part b)

16) a) Relabeling part b)



b) Relabeling part c)



Exercise Group 4.3

- 1) $AB < CD$
- 2) $RS > TU$
- 3) $AC \cong DC'$, $AB < DC'$, $DE < AC$, $AC > DE$, and $DC' > AB$.
- 4) $CD < AX$
- 5) $UT > RV$
- 6) No and no.

Exercise Group 4.4

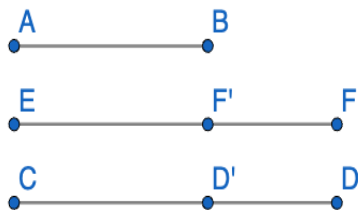
- 1) $AB \cong CB'$ and $CB'D$
- 2) $RS \cong TR'$ and $TR'U$
- 3) Congruence and between
- 4) No and no.
- 5) Yes; $AB \not\cong CD \rightarrow AB > CD$ is false; $AB \cong CD \rightarrow AB \not> CD$ is true

- 6) Nothing. we can not say ADB .
- 7) No, we could have $AB \cong CD$.
- 8) Yes.
- 9) Prove that if $AB < EF$ and $EF \cong CD$, then $AB < CD$. State the converse and contrapositive of this theorem.

Proof. If $AB < EF$ and $EF \cong CD$, then $AB < CD$.

1. Assume $AB < EF$ and $EF \cong CD$.
2. Since $AB < EF$ then there exists F' such that $AB \cong EF'$ and $EF'F$.
3. Let D' be on the same side of C as D such that $EF' \cong CD'$.
4. Since $EF \cong CD$, $EF' \cong CD'$, and $EF'F$, then by Thm 4-3 we are given that $CD'D$.
5. Then $AB \cong EF' \cong CD'$ and $CD'D$ gives us that $AB < CD$.

□



Converse: If $AB < CD$, then $AB < EF$ and $EF \cong CD$.

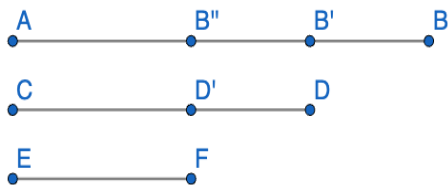
Contrapositive: If $AB \not< CD$, then $AB \not< EF$ or $EF \not\cong CD$.

- 10) Prove that if $AB > CD$ and $CD > EF$, then $AB > EF$.

Proof. If $AB > CD$ and $CD > EF$, then $AB > EF$.

1. Assume that $AB > CD$ and $CD > EF$, which is equivalent to $EF < CD$ and $CD < AB$ by Thm 4-2.
2. Since $EF < CD$, then $\exists D'$ s.t. $EF \cong CD'$ and $CD'D$.
3. Since $CD < AB$, then $\exists B'$ s.t. $CD \cong AB'$ and $AB'B$.
4. Let B'' on the same side of A as B' s.t. $AB'' \cong CD'$.
5. Since $AB' \cong CD$, $AB'' \cong CD'$, and $CD'D$, then by Thm 4-3 we get $AB''B'$.
6. Since $AB''B'$ and $AB'C$ then we know $AB''C$, and since $EF \cong CD' \cong AB''$ then $EF \cong AB''$.
7. Then $EF \cong AB''$ and $AB''C$ gives us that $EF < AB$.
8. Therefore, by Thm 4-2, $AB > EF$.

□



- 11) Prove that if $AB < CD$ and $CD < EF$, then $AB < EF$.

Proof. If $AB < CD$ and $CD < EF$, then $AB < EF$.

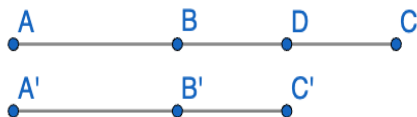
1. Assume $AB < CD$ and $CD < EF$, which is equivalent to $EF > CD$ and $CD > AB$ by Thm 4-2.
2. Then by the previous problem we get $EF > AB$.
3. Therefore, by Thm 4-2, $AB < EF$.

□

- 12) Prove that if B is between A and C , and B' is between A' and C' , and $AB \cong A'B'$, and $BC > B'C'$, then $AC > A'C'$.

Proof. If ABC , $A'B'C'$, $AB \cong A'B'$, and $BC > B'C'$, then $AC > A'C'$.

1. Let ABC , $A'B'C'$, $AB \cong A'B'$, and $BC > B'C' \iff B'C' < BC$.
2. Since $B'C' < BC$ then $\exists D$ s.t. $B'C' \cong BD$ and BDC .
3. Since ABC and BDC , then ADC .
4. Since $A'B' \cong AB$ and $B'C' \cong BD$, then $A'C' \cong AD$.
5. Since $A'C' \cong AD$ and ADC , then $A'C' < AC$. □



- 13) No.

- 14) Prove that if ABC and $A'C'B'$ and $AC \cong A'C'$, then $AB < A'B'$.

Proof. If ABC , $A'C'B'$, and $AC \cong A'C'$, then $AB < A'B'$.

1. Let ABC , $A'C'B'$, and $AC \cong A'C'$.
2. Since $AB \cong AB$ and ABC , then $AB < AC$.
3. Since $AC \cong A'C'$ and $A'C'B'$, then $AC < A'B'$.
4. Since $AB < AC$ and $AC < A'B'$, then $AB < A'B'$. □



Chapter 4 Algebra Review

- 1) Since $0 = -\frac{3}{4} + \frac{3}{4}$, then $0 > -\frac{3}{4}$.
- 2) Since $-\frac{3}{4} = 0 + \frac{3}{4}$, then $-\frac{3}{4} < 0$.
- 3) *Proof.* If $a < b$ then $b > a$.
 1. Let $a < b$, and observe that there must exist some positive number c s.t. $a = b - c$.
 2. Then $a + c = b - c + c \implies a + c = b + 0 \implies a + c = b \implies b = a + c$.
 3. Therefore $b > a$. □
- 4) *Proof.* If $a = b + 1$ and $b = c + 2$, then $a > c$.
 1. Let $a = b + 1$ and $b = c + 2$.
 2. Then $a = (c + 2) + 1 = c + 3$.
 3. Since 3 is positive, then $a > c$. □
- 5) *Proof.* If $a > b$ and $b > c$, then $a > c$.
 1. Let $a > b$ and $b > c$, and observe that there must exist positive numbers d and e s.t. $a = b + d$ and $b = c + e$.
 2. Then $a = b + d = (c + e) + d = c + (e + d)$.
 3. Since $(e + d)$ is positive, then $a > c$. □
- 6) *Proof.* If $a = b - 2$ and $b = c - 3$, then $a < c$.
 1. Let $a = b - 2$ and $b = c - 3$.
 2. Then $a = (c - 3) - 2 = c - 5$.
 3. Since 5 is positive, then $a < c$. □

7) *Proof.* If $a < b$ and $b < c$, then $a < c$.

1. Let $a < b$ and $b < c$, and observe that there must exist positive numbers d and e s.t. $a = b - d$ and $b = c - e$.
2. Then $a = b - d = (c - e) - d = c - (e + d)$.
3. Since $(e + d)$ is positive, then $a < c$. □

8) *Proof.* If $a = b + 3$, then $3a > 3b$ and $-2a < -2b$.

1. Let $a = b + 3$, and observe that $3a = 3(b + 3) = 3b + 9$.
2. Since 9 is positive, $3a > 3b$.
3. Also observe that $-2a = -2(b + 3) = -2b - 6$.
4. Since 6 is positive, $-2a < -2b$. □

9) Prove that if $a > b$, then $ca > cb$ if c is positive and $ca < cb$ if c is negative.

Proof. If $a > b$ and c is positive, then $ac > ab$.

1. Let $a > b$, let c be positive, and observe that there exists a positive number d s.t. $a = b + d$.
2. Then $ac = (b + d) \cdot c = bc + dc$, where d and c are both positive.
3. Since dc is positive, then $ac > bc$. □

Proof. If $a > b$ and c is negative, then $ac < ab$.

1. Let $a > b$, let c be negative, and observe that there exists a positive number d s.t. $a = b + d$.
2. Then $ac = (b + d) \cdot c = bc + dc = bc - (-dc) = bc - d(-c)$, where d and $-c$ are positive.
3. Since $d(-c)$ is positive, then $ac < bc$. □

10) Prove that if $a < b$, then $ca < cb$ if c is positive and $ca > cb$ if c is negative.

Proof. If $a < b$ and c is positive, then $ac < ab$.

1. Let $a < b$, let c be positive, and observe that there exists a positive number d s.t. $a = b - d$.
2. Then $ac = (b - d) \cdot c = bc - dc$, where d and c are both positive.
3. Since dc is positive, then $ac < bc$. □

Proof. If $a < b$ and c is negative, then $ac > ab$.

1. Let $a < b$, let c be negative, and observe that there exists a positive number d s.t. $a = b - d$.
2. Then $ac = (b - d) \cdot c = bc - dc = bc + (-dc) = bc + d(-c)$, where d and $-c$ are positive.
3. Since $d(-c)$ is positive, then $ac > bc$. □